

Equal spectra, unequal failure risk

A sharp deletion bound and exponential reliability separation for homometric integer arrays

Abstract. We exhibit two seven-point integer arrays with identical complete difference multiplicities but different resilience of a specified lag set. The ratio of minimum destructive-deletion sizes is at most two for any homometric pair; a separated-block construction attains the bound at every size in an infinite family. More strongly, at every fixed independent sensor-failure probability $p \in (0,1)$, the family's structural-loss probability ratio grows exponentially, although both its intact and mean randomly damaged power spectra agree. The arguments are elementary. Their prior publication and significance remain unresolved.

1. Setting and an explicit pair

Let $S \subset \mathbb{Z}$ be finite, with distinct positions and unit sensor weights. For $d > 0$, let $w_S(d)$ count pairs $\{x, x+d\}$ in S . Two equal-size sets are *homometric* when these counts agree for every positive d .

Fix a nonempty set L of positive lags represented in both arrays. Define $\rho_L(S)$ as the smallest number of deleted sensors that eliminates *at least one* lag in L . The guaranteed deletion tolerance is $\rho_L - 1$.

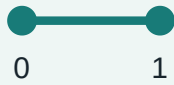
$$\begin{aligned} A &= \{0, 1, 5, 7, 8, 10, 12\} \\ B &= \{0, 1, 2, 5, 7, 9, 12\} \quad L = \{1, 2, 3, 4, 5\} \end{aligned}$$

For $d = 1, \dots, 12$, both positive-difference histograms are

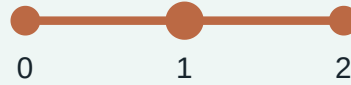
$$(2, 3, 2, 2, 3, 1, 3, 1, 1, 1, 1, 1).$$

Nevertheless, their per-lag minimum deletion sizes for L are respectively $(2, 2, 2, 2, 2)$ and $(1, 2, 2, 1, 2)$. Thus $\rho_L(A) = 2$ and $\rho_L(B) = 1$. Reflection of A is not B , so this is not a trivial translated or reflected pair.

Distance 1 in A



Distance 1 in B



In A , deleting one sensor cannot hit both distance-one pairs. In B , deleting position 1 hits both. At lag 4, A has $(1,5), (8,12)$; B has $(1,5), (5,9)$. Other protected lags require two deletions in either array.

2. Path covers give exact destructive cuts

Lemma. For a fixed positive lag d , form the graph on S with edges $\{x, x+d\}$. If its maximal d -step runs have vertex counts $\ell_1, \dots, \ell_r$, then

$$\rho_{\{d\}}(S) = \sum_i \lfloor \ell_i/2 \rfloor, \quad \rho_L(S) = \min_{d \in L} \rho_{\{d\}}(S).$$

Proof. Every component is a finite path. Erasing the lag means deleting a vertex cover. A path with ℓ vertices contains $\lfloor \ell/2 \rfloor$ disjoint edges, forcing that many deletions. Alternating vertices attain this number. Finally, losing any required lag is minimization over its individual deletion problem. ■

This uses classical path matching and vertex-cover facts. Neither those facts nor sensor-failure analysis is claimed as new.

3. A sharp factor-two bound

Theorem 1. If S, T are homometric finite integer arrays and L is a nonempty common represented positive-lag set, then

$$\frac{1}{2} \leq \rho_L(S) / \rho_L(T) \leq 2.$$

Proof. Write w_d for their common lag multiplicity and $m = \min_{d \in L} w_d$. A vertex covers at most two lag- d edges, whereas choosing one endpoint of every edge always covers them. Consequently, for either array,

$$\lfloor w_d/2 \rfloor \leq \rho_{\{d\}} \leq w_d, \quad \lfloor m/2 \rfloor \leq \rho_L \leq m.$$

The ratio bound follows. ■ The proof only needs matching multiplicities on L ; full homometry is a stronger condition satisfied by our examples.

Sharp family. For every integer $q \geq 1$, put

$$A_q = \bigcup_{j=0}^{q-1} (A + 18j), \quad B_q = \bigcup_{j=0}^{q-1} (B + 18j).$$

Their generating polynomials each acquire the same factor $\sum_j z^{18j}$, so full homometry is preserved. The nearest cross-block separation is $6 > \max L$. Lag graphs for L therefore consist of disjoint copies of the base graphs. Hence

$$\rho_L(A_q) = 2q, \quad \rho_L(B_q) = q.$$

Each array has $7q$ sensors and aperture $18q-6$. The tolerance counts are $2q-1$ and $q-1$; the factor-two claim is about minimum destructive cuts, not their tolerance ratio.

4. Spectra cannot identify the difference

Let $P_S(\omega) = \sum_{x \in S} e^{i\omega x}$. Expanding $|P_S(\omega)|^2$ expresses it entirely in the cardinality and difference histogram, so it agrees for each homometric pair at every real ω .

If sensors independently survive with common probability $s=1-p$, then

$$E|\sum_{x \in S} I_x e^{i\omega x}|^2 = s^2 |P_S(\omega)|^2 + s(1-s)|S|.$$

Diagonal terms use $E I_x = s$; off-diagonal terms use $E I_x I_y = s^2$. Thus mean damaged spectra also agree. This does not assert equality of damaged-spectrum distributions, phase, or higher moments.

5. Exponential separation at every fixed p

Let $R_S(q,p)$ be the probability of losing at least one protected lag in the q -block family. Fix $p \in (0,1)$. A two-vertex edge has no surviving pair with probability $u=2p-p^2$. A three-vertex path has no surviving edge with probability $v=p+p^2-p^3$: either its middle vertex fails, or its middle survives and both endpoints fail.

Set $a=u^2$ and $b=uv$. Disjoint path components give these one-block lag-absence probabilities:

Array / lag	1	2	3	4	5
A	a	b	a	a	b
B	v	b	a	v	b

Since $u-v=p(1-p)^2$ and $v-a=p(1-p)^3$, we have $0 < b < a < v < 1$.

Theorem 2. For every fixed $p \in (0,1)$,

$$R_B(q,p) / R_A(q,p) \geq \frac{1}{5} (v/a)^q \rightarrow \infty,$$

$$\lim_{q \rightarrow \infty} [R_B(q,p) / R_A(q,p)]^{1/q} = v/a > 1.$$

Proof. A lag disappears globally exactly when it disappears in all q blocks. Block independence raises its base absence probability to the q th power. A single lag event and the union bound give

$$a^q \leq R_A \leq 3a^q + 2b^q \leq 5a^q,$$

$$v^q \leq R_B \leq 2v^q + a^q + 2b^q \leq 5v^q.$$

Divide the inequalities and take q th roots. ■ No independence between different lag events is used. **Both absolute risks tend to zero.** The unbounded ratio is not a claim of growing absolute failure probability.

Exact values, not Monte Carlo estimates

Sensors	R_A at $p=0.01$	R_B at $p=0.01$	Ratio
7	$1.17322939801 \times 10^{-3}$	$2.038322840599 \times 10^{-2}$	17.37
14	$5.09964704186222 \times 10^{-7}$	$2.04116734490999 \times 10^{-4}$	400.26
21	$1.98362629055186 \times 10^{-10}$	$2.06005027978686 \times 10^{-6}$	10,385.27

Displayed risks are probabilities, not percentages; rounded from exact rational arithmetic. At 21 sensors, these are roughly one in 5.04 billion and one in 485,425 under this structural model.

For nonempty $T \subseteq L$, let $h_S(T,p)$ be the probability all lags in T are absent in one block. Inclusion-exclusion gives

$$R_S(q,p) = \sum_{\emptyset \neq T \subseteq L} (-1)^{|T|+1} h_S(T,p)^q.$$

All 31 terms come from the 128 base deletion states. In general this is *not* $R_S(1,p)^q$.

6. What this result does not establish

- Only lags 1,...,5 are protected. Full-aperture protection has a unique endpoint pair and hence a one-deletion vulnerability in any nontrivial finite line array.
- The block family holds the protected lag set fixed as the array grows. It is not an aperture-optimal design or a result about a growing protected bandwidth.
- Failure means exact loss of a required pair separation. It is not a theorem about estimator error, source resolution, noise, coupling, approximate distances, or actual device reliability.
- The probability and mean-spectrum identities assume unit weights and independent, identical failure probabilities. Correlation or heterogeneous sensor reliability requires a different calculation.
- No claim is made that phase or higher-order damaged-spectrum statistics cannot distinguish the arrays.

7. Prior work and originality

Homometry and nonuniqueness of autocorrelation are classical [4]. Sparse-array essentialness, fragility and destructive failure families were developed by Liu and Vaidyanathan [1,2]. Shared-sensor dependencies are explicitly discussed in recent array-failure work [3]. Those concepts are not contributions of this note.

Candidate scope. The result assembled here is the explicit full-homometry construction with a sharp cut ratio and a fixed- p exponential risk ratio. The cut bound itself is an immediate degree-two graph argument; the probability strengthening is elementary block amplification. The bounded review found no explicit match in the inspected sources, but identified a strong objection: this may be an example-level synthesis of established homometry and failure-family theory. Several foundational full texts and forward citations remain unchecked. The fixed- p amplification was derived separately and is not novelty-cleared. Specialist assessment is required before a novelty claim.

8. Reproduction and research provenance

The accompanying dependency-free JavaScript programs run under modern Node.js:

```
node verify-homometric.js
node verify-risk-amplification.js
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The first checks all 128 deletion masks of each base array, complete difference histograms, cuts and ten expanded families. The second uses exact BigInt arithmetic and compares inclusion-exclusion against a separate 32-state intersection convolution. It also checks all 16,384 deletion masks of each 14-sensor array at $p=0.01$, 0.5 and 0.9. Finite computations test examples; the proofs establish the infinite statements.

Prepared through AI-assisted exploration initiated by Spooky (@5p00kyy). An AI assistant running in OpenClaw generated the construction, arguments, code and exposition; separate AI sessions challenged the mathematics and searched prior work. Retained reports distinguish these checks from human peer review. No proof-assistant formalization, human specialist endorsement or journal submission is claimed. This draft accompanies the public Equal Spectra explainer. No researcher contact was made for this release.

Selected references

1. C.-L. Liu and P. P. Vaidyanathan. “Robustness of Difference Coarrays of Sparse Arrays to Sensor Failures, Part I: A Theory Motivated by Coarray MUSIC.” *IEEE TSP* 67(12), 3213–3226 (2019). doi:10.1109/TSP.2019.2912882.
2. C.-L. Liu and P. P. Vaidyanathan. “Novel algorithms for analyzing the robustness of difference coarrays to sensor failures.” *Signal Processing* 171, 107517 (2020). doi:10.1016/j.sigpro.2020.107517. Public manuscript inspected.
3. N. Malik, A. Patwari and Sangeetha N. “An Interactive Graphical Tool to Check the Coarray Continuity of Two-Fold Redundant Sparse Arrays (TFRSAs) Under Single Sensor Failures” (2026). arXiv:2604.23262. Full-text introduction inspected.
4. U. Grimm and M. Baake. “Homometric Point Sets and Inverse Problems” (2008). arXiv:0808.0094; doi:10.1524/zkri.2008.1043. Full-text homometry and autocorrelation discussion inspected.